

Targeted SAT Math — Set 05

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Homework after Bluebook Practice Test 6 and Suite Hard Module 01. Work on paper.

Calculator / Desmos is fine.

These are **new** questions on the items that missed — not those official questions, and not copies from Sets 01–04.

SPR = student-produced response. Write a number (integer, decimal, or fraction). On #7, interpret $A \mid B$ and $A \cap B$, write the formula, then compute. #8 is a count-then-condition item — no interpretation blanks.

Write the job first.

1. Notes for this set only

Polynomial in factor form — the constant c

If the zeros are r, s, t , a monic cubic is

$$h(x) = (x - r)(x - s)(x - t) = x^3 + \cdots + c.$$

The constant c is what you get when you multiply the three constant pieces of those factors. You do **not** need a or b .

- Zeros $-2, -3, 4$ means the factors are $(x + 2)(x + 3)(x - 4)$. The constant is $2 \cdot 3 \cdot (-4)$, not a guess from multiplying the zeros raw.
- Same idea for a degree-6 polynomial: six factors, six constants, multiply.
- If the leading coefficient is **not** 1, multiply that leading number through at the end. c scales with it. Look at the number in front of the highest power of x before you multiply.

Finish the rectangle

If one side is d more than the other and they give a diagonal (or the square of a diagonal), write

$$x^2 + (x + d)^2 = (\text{diagonal})^2$$

and solve. Ugly numbers are not a signal to hunt for a trick. They are a signal to write Pythagorean and finish.

Infinitely many solutions

Two linear expressions in x are the **same line** when matching coefficients are equal: the x -coefficients match, **and** the constants match.

If the right-hand side already shows a constant K , and the question asks for the constant term of the expanded left-hand side, that constant **is** K . You do not have to finish the expansion.

If they ask for the value of p that makes a system have infinitely many solutions, scale one equation until the x and y pieces match, then force the constants to match too. That same p is also the cutoff: any other value of p gives **no** solution (parallel, different constant).

Minimum of a quadratic — which x

For $f(x) = ax^2 + bx + c$ with $a > 0$, the minimum is at

$$x = -\frac{b}{2a}.$$

That is the whole job if they asked for the x -value. Completing the square finds the vertex (h, k) . You do not need k unless they asked for the minimum **value**.

Conditional probability — write the events first

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- If the stem already says what A is and what B is, do **not** rewrite those two. Interpret $A \mid B$ and $A \cap B$ only.
- Event $A \cap B$ is **both** sentences at once. The symbol $\cap$ means “and.”
- Event $A \mid B$ is “ A happens, **restricted to the cases where B already happened.**” The denominator is no longer the whole table. It is only the B slice.
- $P(B)$ is (count of B) / (grand total). $P(A \cap B)$ is (count of both) / (grand total). $P(A \mid B)$ is (count of both) / (count of B).

On #7, write $A \mid B$, $A \cap B$, and the formula **before** you divide.

Some items never hand you a finished table. They give **rows** of two kinds, and each row has two counts. Multiply first: (number of rows) $\times$ (count per row). That fills the table. Then condition. A leftover per-row number is not the probability.

Circle moved, radius changed

$$(x - h)^2 + (y - k)^2 = r^2.$$

- Up p units: k becomes $k + p$. Down p : k becomes $k - p$. Left/right changes h .
- A new radius R means the right-hand side is R^2 , not the old r^2 .
- Write the new equation. Do not reuse the old radius just because the center movement was the part you noticed first.

Factored form $\rightarrow a + b$ or $a + 2b + 3c$

Zeros at r and s means $f(x) = a(x - r)(x - s)$. One extra point finds a . Then expand to $ax^2 + bx + c$ and evaluate the combination they asked for.

$a + b + c = f(1)$. $a + 2b + 3c$ is **not** $f(1)$. Expand, then plug the three coefficients into that expression.

Tangents around a circle, then a right triangle

- Two tangent segments from the **same exterior point** are congruent.
- The radius to a point of tangency is **perpendicular** to that tangent. That is a right triangle: radius, tangent segment, line from the center to the exterior point.
- A huge number is not a reason to abandon Pythagorean. Write $r^2 + t^2 = d^2$ and compute.
- One SAT kite is **two radii plus two tangents**. Let each tangent be x . Perimeter is $2r + 2x$. Solve for x . If the asked length is the line from the center to the exterior point, that length d satisfies the quadratic $d^2 = r^2 + x^2$.

2. Questions

1. (SPR) The function h is defined by $h(x) = x^3 + ax^2 + bx + c$, where a , b , and c are constants. The zeros of h are -5 , -6 , and -7 . What is the value of c ?

2. (SPR) The function p is defined by $p(x) = x^6 + ax^5 + bx^4 + dx^3 + ex^2 + fx + c$, where the letters other than x are constants. The zeros of p are $-1, -1, -2, -3, 3$, and 1 . What is the value of c ?

3. (SPR) The function h is defined by $h(x) = \textcircled{2}x^3 + ax^2 + bx + c$, where a, b , and c are constants. The zeros of h are $-5, -6$, and -7 — the same three zeros as in question 1. What is the value of c ?

The leading coefficient is the circled number. It is not 1.

4. (SPR) In rectangle $PQRS$, side $\overline{PQ}$ is 12 units longer than side $\overline{PS}$. The square of the length of diagonal $\overline{PR}$ is 4,304. What is the length of the longer side of the rectangle?

5. (SPR) The equation $11p(x - 5) + 3(x + 7) = sx + 214$ has infinitely many solutions, where p and s are constants. What is the value of $3(7) - 55p$?

6. (SPR) The function f is defined by $f(x) = 2x^2 - 11x + 15$. What is the value of x at which f takes its minimum value?

7. A grove has only spruce trees and elm trees. Each tree is tagged or not tagged. The table shows the counts.

	Tagged	Not tagged	Total
Spruce	14	6	20
Elm	9	21	30
Total	23	27	50

A tree is selected at random. Let A be the event that the tree is a spruce. Let B be the event that the tree is tagged.

Interpret event $A \mid B$ and event $A \cap B$. Then write the formula, then the probability.

- Event $A \mid B$: _____

- Event $A \cap B$ (the $\cap$ symbol means “and”):

- Formula: $P(A \mid B) =$ _____
- (SPR) The value of $P(A \mid B)$: ____

8. A farm has 4 rows of watermelon plants and 6 rows of blueberry bushes. Each watermelon row has 7 plants that produced ripe fruit and 5 plants that did not. Each blueberry row has 10 bushes that produced ripe fruit and 3 bushes that did not. A plant or bush from one of these rows will be selected at random. What is the probability of selecting a watermelon plant, given that the selected plant or bush produced ripe fruit?

- A) $\frac{7}{12}$
 B) $\frac{22}{15}$
 C) $\frac{22}{14}$
 D) $\frac{63}{63}$

9. (SPR) In the xy -plane, a circle has equation $(x + 4)^2 + (y - 1)^2 = 49$. The circle is translated 6 units down, and the radius of the resulting circle is 3. Write the equation of the new circle as $(x - h)^2 + (y - k)^2 = r^2$. What is the value of $r^2 - k$?

10. (SPR) A quadratic function f has zeros at $x = -3$ and $x = 5$, and the graph of $y = f(x)$ passes through $(0, 30)$. If $f(x) = ax^2 + bx + c$, what is the value of $a + 2b + 3c$?

11. (SPR) A circle with center C has radius 1,365. Four lines, each tangent to the circle, form quadrilateral $HXYZ$. The circle lies inside the quadrilateral and is tangent to all four sides. Point H is a vertex of the quadrilateral — it is the intersection of the two tangent lines that contain sides $\overline{HX}$ and $\overline{HZ}$.

The perimeter of $HXYZ$ is 13,312. Side $\overline{XY}$ has length 4,056, side $\overline{YZ}$ has length 4,056, and $\overline{HX} = \overline{HZ}$.

The circle is tangent to $\overline{HX}$ at point P , and $PX = 780$.

The radius to a point of tangency is perpendicular to the tangent line. What is the distance CH ?

12. (SPR) In the system below, p is a constant. The system has infinitely many solutions for one value of p , and no solution for every other value of p . What is the value of p that gives infinitely many solutions?

$$6x + 3y = p$$

$$2x + y = 4$$

13. (SPR) A circle has center C , and points A and B lie on the circle. Segments $\overline{AD}$ and $\overline{BD}$ are tangent to the circle at A and B , respectively. The radius of the circle is 324. The perimeter of quadrilateral $CADB$ is 3,528. What is the distance CD ?

Let the unknown first. The two tangent segments are equal. The two radii are equal. Radius to a point of tangency is perpendicular to the tangent.